Subject Description Form

Subject Code	**ITC3027T**
Subject Title	**Fabric Technology I**
Credit Value	**3 credits**
Level	**3**
Pre-requisite/ <Co-requisite> / (Exclusion)	ITC2004T Textile Studies II
Objectives	The subject aims to further reinforce students' understanding of both knitted and woven fabric in terms of formation and characteristics. This subject also extends the scope of studies in order to provide the student with a more comprehensive knowledge in fabric technology as required by a textile technologist.
Intended Learning Outcomes	Upon completion of the subject, students will be able to: a. describe the use of circular knitting in the fashion industry; describe the classification of circular knitted fabrics; b. describe the use of straight bed knitting machine in the fashion industry; analyze and compare the warp knitted fabrics using different yarn, guide bar threading and lapping movements; c. Identify fabrics produced by different shedding systems; compare different weft insertion systems and their applications; d. Analyze woven fabric design from colour and weave effect, advanced weaves; e. Communicate efficiently and clearly with textile professionals
Subject Synopsis/ Indicative Syllabus	**(I)** **Circular knitting:** structures and knitting of some common single and double jersey circular knitted fabrics **(II)** **Flat knitting:** string formation knitting, applications of loop transfer. **(III)** **Warp knitting:** Tricot and Raschel machines, fabrics from part set guide bars **(IV)** **Different shedding systems:** Fabrics such as Jacquard, small patterned by Dobby and basic weaves by Cam/Tappet shedding **(V)** **Different weft insertion systems:** Comparison in terms of Productivity, variety, cost and performance **(VI)** **Advanced fabrics and color-weave effect**: Double clothes, pile fabrics and some classical color-weave effect patterns.
Teaching/Learning Methodology	Teaching will be primarily conducted in lectures, which are supplemented by tutorials. Appropriate demonstration of the fabric samples and analysis methods, fabric production machineries and related tools and instruments will be conducted in the weaving and knitting laboratories.

<table>
<tr><td>Assessment Methods in Alignment with Intended Learning Outcomes</td><td>
<table>
<tr><td rowspan="2">Specific assessment methods/tasks</td><td rowspan="2">% weighting</td><td colspan="6">Intended subject learning outcomes to be assessed (Please tick as appropriate)</td></tr>
<tr><td>a</td><td>b</td><td>c</td><td>d</td><td>e</td><td></td></tr>
<tr><td>1. Coursework</td><td>50%</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td><td></td></tr>
<tr><td>2. Final Examination</td><td>50%</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td><td></td><td></td></tr>
<tr><td>Total</td><td>100%</td><td colspan="6"></td></tr>
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Explanation of the appropriateness of the assessment methods in assessing the intended learning outcomes:
This subject provides students with lectures and laboratory training. Courseworks provide an effect assessment of the progress of the student especially of the part laboratory training. A final examination is appropriate to assess all five learning outcomes.
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<tr><td>Student Study Effort Expected</td><td>
<table>
<tr><td>Class contact:</td><td></td></tr>
<tr><td>▪ Lecture</td><td>22 Hrs.</td></tr>
<tr><td>▪ Tutorial</td><td>4 Hrs.</td></tr>
<tr><td>▪ Laboratory</td><td>18 Hrs.</td></tr>
<tr><td>Other student study effort:</td><td></td></tr>
<tr><td>▪ Assignments</td><td>59Hrs.</td></tr>
<tr><td>Total student study effort</td><td>103 Hrs.</td></tr>
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<tr><td>Reading List and References</td><td>
Spencer, D., Knitting Technology, 3rd Edition, Woodhead Publishing Ltd.
Raz, Smauel, (1993). Flat Knitting Technology, Universal Maschienfabrik,.
Wilkens, Chris, (1997). Warp Knit Machine Elements: All About Tricot-, Raschel- and Crochet Warp Knitting Machines as well as Stitch Through Mali-machines, U. Wilens Verlag.
Lord & Mohamed, (1982). Weaving: Conversion of Yarn to Fabric, 2nd Ed., Merrow.
Mark & Robinson, (1976). Principles of Weaving. Textile Institute.
Robinson, A. T. C and Marks, R. (1973). Woven Cloth Construction. The Textile Institute.
Walker, R.P. and Knight , W.C., Fabric Designing Fundamentals.
<u>Adanur, Sabit, </u>(2001). Handbook of Weaving: Technomic Pub. Co. Inc.
Z. J. Grosicki, Watson's, (1977). Advanced Textile Design: Compound Woven Structures, London ; Boston : Newnes-Butterworths.
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